Term Project:
Socket Programming

Group X

6xxxxxxxx21 Name Surname

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Part 1 System Architecture Design

This part must be fully completed; otherwise, Part 2 will not be graded.

R1 – System Architecture Design



- Explanation

R1 – System Architecture Design

- Explain the computer network protocols used in the project
  - What protocols are used
  - Why each protocol was selected
  - How each protocol is used within the project

R2 – The chat messages between the server and each client must be implemented using Socket Programming only

Code Snapshot

Generative AI Prompt (if use)

Part 2 Fundamental implementation

This part must be fully completed; otherwise, Part 3 will not be graded.

R3 – Each client must have a unique name.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R4 – Each client can see a list of names of all clients that are currently connected to the server including its own name

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R5 – Chat between clients (both Private and Group message) must have its own chat room

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R6 – Each chat room (both Private and Group message) must include a chat box for sending text messages and chat window for displaying the sent messages and new incoming messages.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R7 – Each client can send a direct text message to any clients in the list. Only the sender and receiver can see the messages.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R8 – Each client can create a chat group(s), which initially includes only themselves as a member.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R9 – Each client can see a list of all existing chat groups (with the member list) created by any clients.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R10 – Clients can choose to join a group chat only by themselves; they are not added automatically by the group creator.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

R11 – Each client can send a text message to the group that they joined. Only the members of the chat group can see the messages.

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

Part 3 Special Points

At least one of the features in the project must be an AI feature, such as speech-to-text, text summarization, sentiment analysis, a chatbot, etc., to demonstrate the integration of AI technologies into your system.

Special Point 1: (AI feature) ..... (Description)

AI Model : ..... (ref. .....)

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

Special Point 2: ..... (Description)

Code Snapshot (Mandatory)

Generative AI Prompt (if use)

Code Snapshot

Generative AI Prompt (if use)

Special Point 3: ..... (Description)

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)

Special Point 4: ..... (Description)

Code Snapshot (Mandatory)

Code Snapshot

Generative AI Prompt (if use)

Generative AI Prompt (if use)